DSTM Airport Simulation Communication Counts

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1 Introduction/Detour

If you are interested in an explanation of the simulation and our DSTM vs MPI versions, those details are located in (*/dstm/examples/airport_sim/scripts/scales*). This report is explicitly detailing the timings of MPI communication calls used in both the DSTM and MPI versions of this simulation.

2 Experiment

We run a weak scaling experiment. We log the timings of MPI communications used throughout the simulation. This logs were placed directly within the source code of both DSTM and the MPI airport simulation. Therefore, the *dstm_log()* calls may or may not be present within the github repository. In the DSTM example, these specific logs bloat our log files greatly.

Configuration.

- Airports per rank (fixed): 32
- Timesteps per run: 8
- Per-arrival synthetic work: 1,000 iterations
- Rank sweep: $P \in \{1, 2, 4, 8, 16, 32, 64, 128, 256\}$
- Total airports (varies): $32 \cdot P$, from 32 at P=1 to 8192 at P=256
- Nodes (Cobalt allocation): 8
- Ranks-per-node: $\max(1, P/8)$, distributed via `--map-by ppr:PPN:node`

Builds compared.

- `dstm`: original DSTM-backed communication, at `../airports/airports`.
- `mpi`: faithful MPI port using `Isend`/`Irecv` with per-timestep barrier, at `../mpi_air/airports`.

3 Results

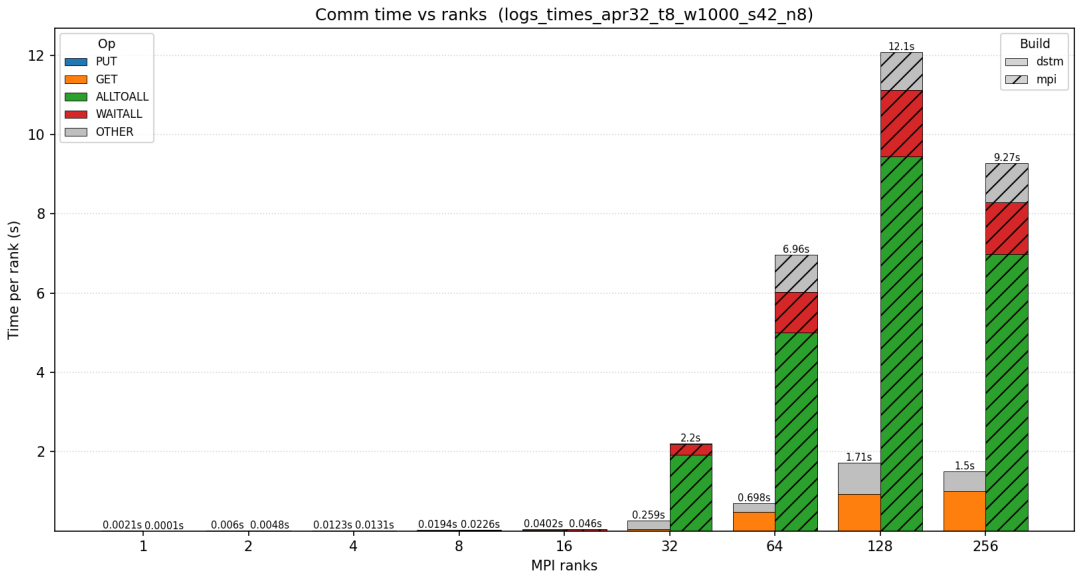


Figure 1: Weak scaling: total application runtime vs. rank count for the DSTM and MPI builds, with the per-rank airport count held fixed at 32. At each rank count the two builds are shown side-by-side; bar height is the end-to-end elapsed time reported by the simulator, and each bar is stacked by the average per-rank time spent in each tracked communication operation (`PUT`, `GET`, `ALLTOALL`, `WAITALL`), with the remaining non-communication time shown as `OTHER`. Shorter bars indicate less wall-clock time to complete the same per-rank workload, and a smaller colored (non-`OTHER`) fraction indicates that less of that runtime was spent in communication.